

PHD DEFENSE · TU DELFT

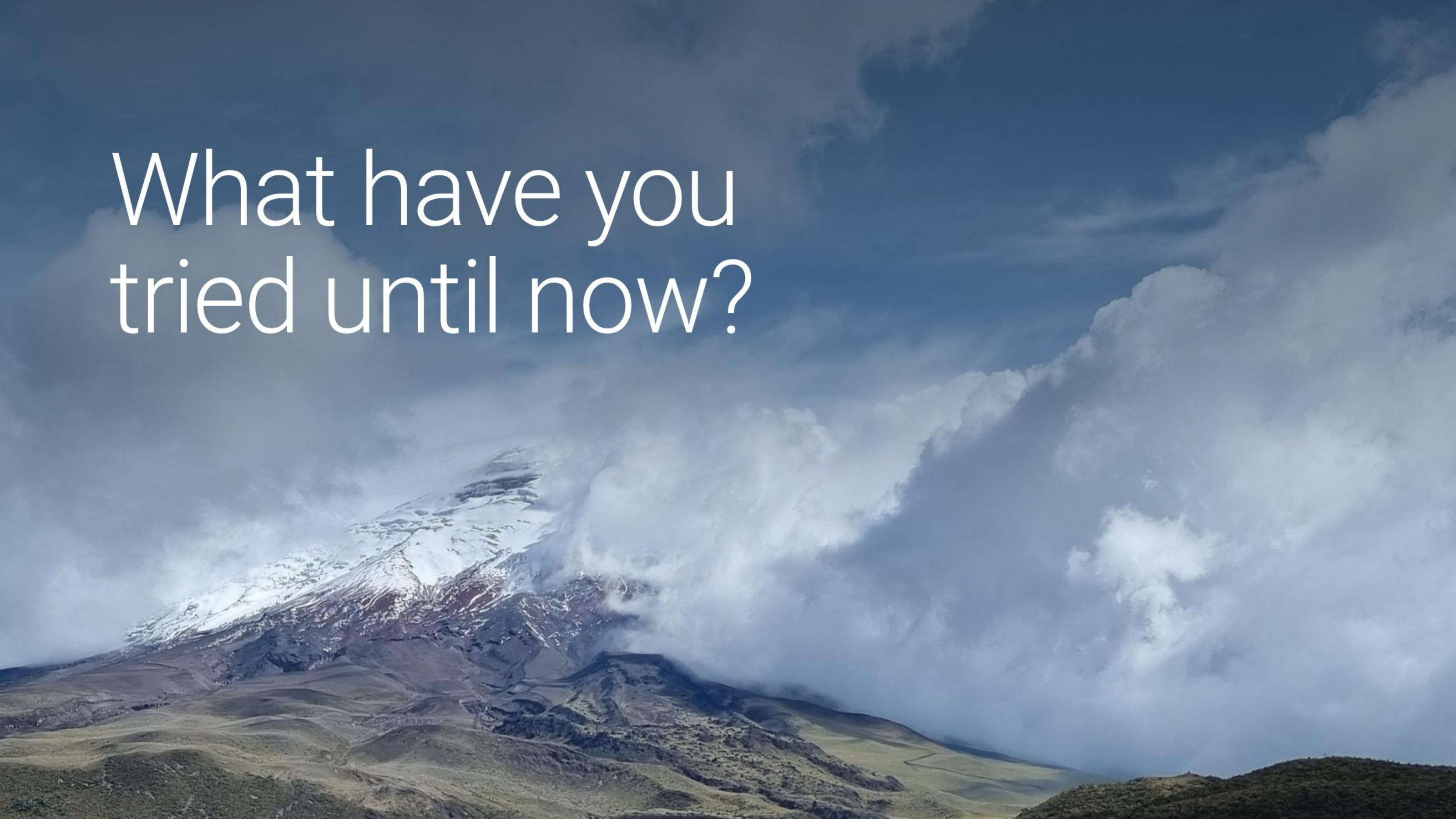
Sequences in Data to Represent Learning Processes

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It's not about arriving first, but knowing how to arrive.

Jose Alfredo Jiménez

What have you
tried until now?



THE PROBLEM, LONG BEFORE AI

We teach the **process**.
We grade the **product**.

IN THE LAB

“What have you
tried until now?”

HOMEWORK, EXAMS, AUTOMATED GRADERS

“Did you get the
right answer?”

EVERYTHING THE SYSTEM RECORDED

WHAT GOT GRADED



THEN AI ARRIVED

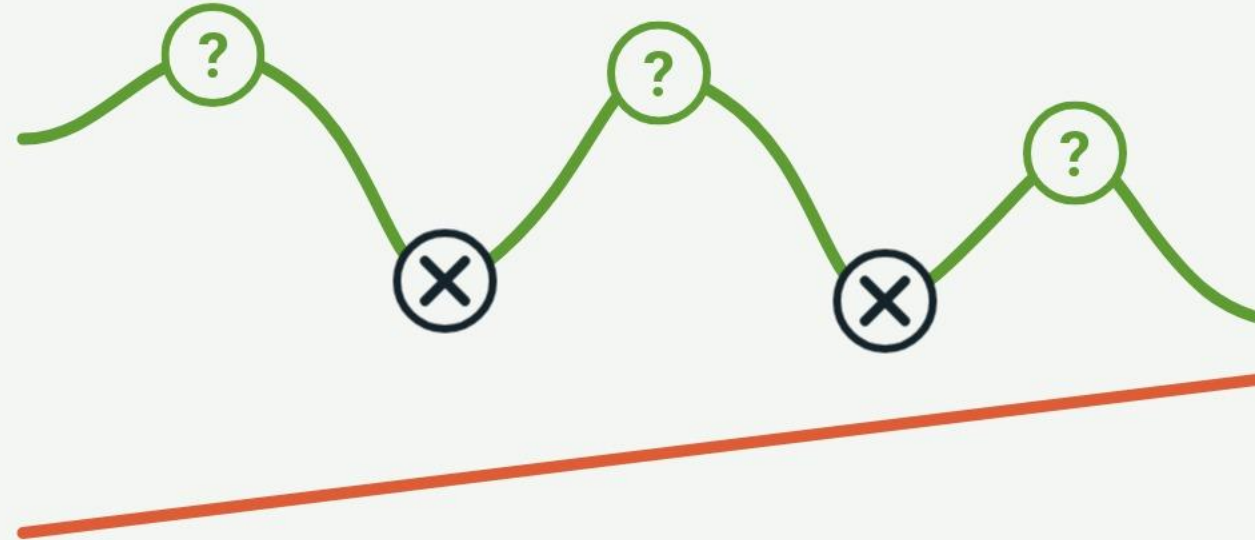
Same homework. Very different learning.

Struggles through it

tries, fails, asks why, tries again

Asks the AI

one prompt, one paste



homework.py

THE QUESTION

If the finished work tells us nothing, what do we look at instead?

The process. We can identify what a student needs, and when.

PHASE 1 OF 3 · LOOK

Can existing data show how students work?

It missed the errors, and it had no AI in it.



PHASE 2 OF 3 · BUILD

A tutor that sees the process as it happens

Built into the notebooks students already use, so it can respond to how they work.



PHASE 3 OF 3 · SUPPORT

Help students ask better questions

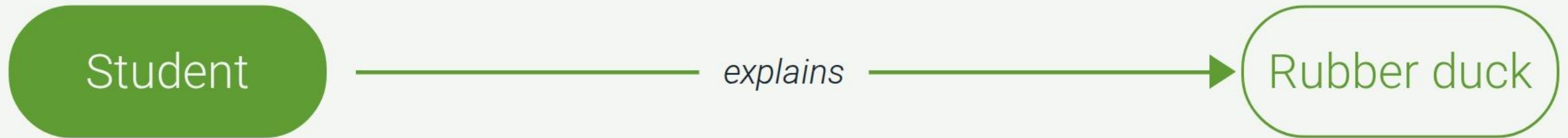
In classrooms and online, while the work is still happening.



WHAT WE SAW

From rubber duck to solution vending machine

BEFORE



Explaining the problem is how you find it.

WITH AI



The answer arrives. The thinking doesn't.

WHAT WE FOUND

Left alone, bad habits with AI don't fix themselves.

Students who started with “fix this for me” kept asking that way, week after week.



THE HOPEFUL PART

We can see these habits,
support them, and change them.

A SHORT LESSON

Changed how students
asked for help

A SMALL ACKNOWLEDGMENT

Noticing a better
question kept them trying

One lever. There are more to find.

Early results, from small groups.

WHAT THIS MEANS

Three things we can do now

TEACHERS

Teach how to ask a good question, not only how to find the answer.

TOOLMAKERS

Design for restraint: a hint before the answer.

INSTITUTIONS

Measure the journey, not only the finished product.

Thank you

When the final answer reveals
nothing about the learning
that occurred to produce it,
the process is all we have left.

